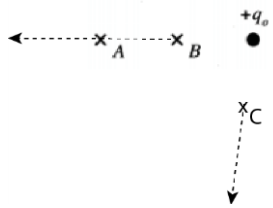
Week 2: Voltage, Capacitors, Circuits and Dipoles SOLUTIONS

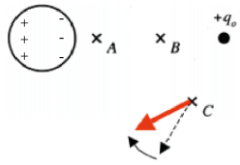
Discussion Questions

1 positive and negative charges equidistant from the point P mostly contribute equal and opposite amounts to the potential, except for the -2q pair, hence -2kq/d + -2kq/d = -4kq/d

2 (a) direction is radially outwards from the charges as shown.



V_{BC} = 0 (same distance), hence $V_{AB} > V_{BC}$ and $V_{AC} = V_{AB}$;
(b) polarization of charges with + to the left of sphere and – to the right; field at A and B both greater; field at points A & B same direction, at C it has shifted slightly due to induced charges on the sphere (see pic below). V_B and V_A have reduced;



3 (a) Field at centre is zero (inside a conductor); potential is equal to the potential at the surface; find this from $V = kQ/r = 4\pi \times 10^9$ V.
(b) Field is still zero (symmetry argument), but the potential will now be larger, as more work will have to be done to bring a charge to the centre of the sphere.

4 (a) C increases (from $C = A\varepsilon_0/d$) (b) PD remains constant because plates are connected to the battery; (c) Q increases because $Q = CV$ and C has increased; (d) E increases since the charge density on the plates has increased (from $E = \rho/\varepsilon_0$)

5 (a) C increases (from $C = A\varepsilon_0/d$) (b) V will reduce because $V = Q/C$ and Q must be the same (c) Q remains constant because plates are disconnected from the battery (d) E will remain the same, since (from Gauss) $E = \rho/\varepsilon_0$ and ρ is the same.

6 E the same (from $E = V/d$); V is the same because it is still connected to the battery; Q increases because C has increased and V has remained the same ($Q = CV$); E increases from $E = 0.5CV^2$.

7 E reduces because of the polarization of the dielectric; V reduces because C has increased and Q the same (charges on plates cannot move; $V = Q/C$); energy reduces (from $E = Q^2/2C$ or $0.5QV$)

8 Resistors in parallel always reduce the total resistance -more paths for current to pass through (try to steer clear of "just follows from formula" reasoning).

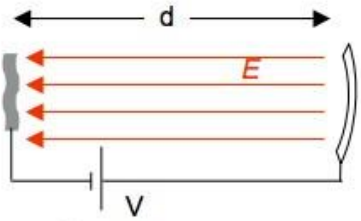
9 Nothing changes because the switch simply connects two points that are at the same potential. No current flows along the link.

10 When the wire is connected, it 'short circuits' bulb B. Bulb A now receives the full 12 V (instead of 6 V previously). A burns more brightly. B goes out.

11 (a) The brightness is an indication of the power. Using $P = V^2/R$, it is clear that the voltage across the bulbs is the easiest way to predict brightness. A will receive two thirds of the battery voltage as its resistance is double that of the two bulbs in parallel. The other answers follow (since the bulbs in the circuit at right are identical and assuming the battery is ideal).
(b) C is now out of the circuit (and out); each remaining bulb receives exactly half the battery voltage.
(c) Student 1 is only partially correct, as is Student 2. The key is that the brightness depends on V^2/R, and as the resistance can (perhaps) be assumed to be the same, one should look at the voltage across the two lamps. The voltage across A decreases, across B increases. So B will brighten and A will get less bright.

Problem Solving Questions

12



$d = 0.5$ m, $\Delta U = 4.8\times10^{-15}$ J

a) The potential difference between filament and screen is the potential energy difference per unit charge, i.e. $\Delta V = \frac{\Delta U}{q}$, with $q = q_{electron} = 1.6\times10^{-19}$ C. $\therefore \Delta V = 30$ kV

b) $E = \frac{\Delta V}{d}$ (if the field is assumed to be uniform)
$= 60$ kV/m directed toward filament

c) $F = qE$
Assuming the field is uniform, the force is the same at the filament and at the screen.
$F = 9.6\times10^{-15}$ N directed toward screen.

13 Torque from field $= \boldsymbol{p} \cdot \boldsymbol{E}$; max magnitude is $pE = 3.1 \cdot 10^{-23}$ Nm; PE given by $-\boldsymbol{p.E}$, hence maximum
change is $2pE = 6.2 \cdot 10^{-23}$ J. .

14 All points on the ring are equidistant from the centre, hence $V = kQ/r = k\rho 2\pi r/r = 2\pi k\rho = 1.41$ V

15

a) $C = \frac{\varepsilon_0 A_1}{d}$

b) The charges are free to move across to new extension to the plates A_2, but we cannot change the total charge Q. Hence the surface charge density is $\sigma = \frac{Q}{A_1+A_2}$
The electric field between the plates is $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0(A_1+A_2)}$. The potential across the plates is the electric field multiplied by the separation, i.e. $V = Ed = \frac{Qd}{\varepsilon_0(A_1+A_2)}$.

c) i) Charge conserved and area unchanged, hence no change in charge density.
ii) Outside the plates, $E = 0$, hence no change.
Inside the plates, $E = \sigma/\varepsilon_0$, hence no change (ignoring any edge effects)
iii) $V = Ed$ and E is constant. Hence V increases proportionally, by a factor D/d.

16 Set the axis scale as follows: $x = 0$ at LH end, $x = L$ at RH end, and $x = L + a$ at P. Break up rod into small segments, width dx. The charge dq on each element $= \lambda\, dx$. Contribution to potential at point P is then $V = \frac{\lambda}{4\pi\epsilon_0}\frac{dx}{L+a-x}$. Integrate from $x = 0$ to $x = L$. This gives: $V = -\frac{q}{4\pi\varepsilon_0 L}\log_e\frac{L+a}{a}$

17 Use $u_E = 0.5\kappa\varepsilon_0 E^2$, then $u_E \times$ volume ($= Ad$) $= 6.6 \times 10^{-5}$ J

18 (a) Energy of 100 W lamp for 8 h is 800 W h; i.e. 400 AA batteries. Hence \$400.
(b) \$0.25 per kW h is 0.025 c per W h; multiply by 800 gives 20c.

19 Kirchhoff's laws: only the voltage loop laws are needed here; they give $5 = I_1R_1$ (small loop) and $4 + 5 - 6 - I_2R_2 = 0$; (Solve to find currents $I_1 = 50$ mA & $I_2 = 60$ mA)

20 In the first circuit the loop law gives $\varepsilon - 0.5 \times R_{int} - 0.5 \times 6 = 0$; the second circuit gives $\varepsilon - 0.8 \times R_{int} - 0.8 \times 3 = 0$. Solve simultaneously for $R_{int} = 2.0\ \Omega$ and $\varepsilon = 4.0$ V

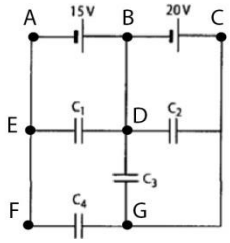
21 Use $C = A\varepsilon_0/d$ and $C = \kappa A\varepsilon_0/d$; also energy stored $= \frac{1}{2}CV^2$; difference is 1.89×10^{-4} J

Past Exam Questions

22

(a) **(i)** E is radially inwards and given by $kq/r^2 = 4500$ V/m
(ii) Use $V = kq/r = -900$ V
(iii) 900 eV

(b) Solutions helped if you annotate the diagram as below:



Clearly, potentials around the circuit are as follows: A = 0 V (reference point), hence E = F = 0 V; B & D = 15 V, C & G = 35 V. Answers follow simply:
C1 = 15 V; C2 = 20 V; C3 = 20 V; C4 = 35 V.

23

• V across both capacitors must remain equal (though less than before, since C has increased for no change in total charge.
• charge before $= 2Q$; charge afterwards $= Q_0 + Q'$; hence $2Q = Q_0 + Q'$ .....(1)
• voltage afterwards the same on both capacitors, hence $Q_0/C = Q'/\kappa C$.....(2)
• Simultaneous equations - eliminate Q_0 and rearrange.
• answer is $Q' = \frac{2\kappa Q}{\kappa+1}$

24

(a) Use $Q = It = 0.5 \times 0.1 \times 10^{-6} = 5 \times 10^{-8}$ C. Divide by charge per electron, $\frac{5\times10^{-8}}{1.6\times10^{-19}} = 3.125\times 10^{11}$electrons .

(b) Use $I = Q/t \rightarrow 500\times\frac{5\times10^{-8}\text{C}}{1\text{ sec}} = 2.5\times10^{-5}$A (c) Average power = 50 MeV × no. of electrons per second; or an electron moving though a 50MeV potential will gain 50MV, so $P_{avg} = I_{avg}V = (2.5\times10^{-5}\text{A})(50\times 10^{6}\text{V}) = 1250$ W. Peak power = 50 MeV × no. of electrons in a pulse divided by 0.1×10^{-6}. So, $P_{\text{peak}} = 1.56$ MeV/s [*will have to be converted into joule (1.6×10^{-19}* J per eV)]; or $P_{peak} = I_{peak}V = (0.5\text{A}\times50\text{MV}) = 25$ MW

25

a) Junction equation: $I_1 + 2 = I_2$
K. Loop equation #1: $10 - 8I_1 + 2\times3 = 0$
K. Loop equation #2: $\varepsilon - 2I_2 - 3\times2 = 0$
Solve to find either I_1 or I_2:
$I_1 = 2\text{ A} \rightarrow \varepsilon = 14\text{V}$

b) No – while the globe's resistance in this configuration can be determined by V=IR, when different voltages or currents are applied, R may change. Ohm's Law defines an ohmic device to have a constant R for any voltage or current. Further tests would be required to confirm this statement.

26a) In the following solution, current I1 flows through R1 and R2, and current I2 flows through R3 and R4. No current flow in the ammeter means the current through R_1 = current through R_2, and current through R_4 = current through R_3. (The change in potential or voltage across each half of the diamond must be the same, so that no potential difference exists between A and B to drive current flow.)From the loop rule passing through R_1, point A and R_4, we get the relationship $I_1R_1 = I_2R_4$ and similarly $I_1R_2 = I_2R_3$. Dividing one of these equations by the other leads to the required result. These values satisfy the relationship described in part (a). The voltage drop over R_1+R_2 = R_4+R_3 will be 5.0 V. By symmetry/ voltage divider rule, voltage decrease across R_4 = R_1 will be 3.0 V. So $V_A = V_B = 2.0$V